

【8.24| The Guardian】 Incredible ways **insects** are communicating and listening all around us **昆虫交流与倾听的奇妙方式**

难度评级：★★★★☆

【考研那些事儿之自然系列】

今日文章脉络：

我们身边充斥着人类听不懂的生物"对话" (Para. 1-2)



蜘蛛通过调节蛛网振动来"听"辨猎物 (Para. 3)



信息素——无脊椎动物的化学语言 (Para. 4)



拟态——用于欺骗的交流工具 (Para. 5)



蝴蝶幼虫用双重"歌声"在蚁巢中寄生生存 (Para. 6)



交流不限于无脊椎动物之间，植物与昆虫共生 (Para. 7-8)



无脊椎动物交流的进化本质：合作与竞争并存 (Para. 9)



人类活动干扰昆虫交流，最终危害人类自身 (Para. 10)

背景知识：昆虫的信息素与拟态交流

<https://www.nationalgeographic.com/animals/article/insects-communication-pheromones>

【学习流程：第一遍请先看原文自己精读，第二遍对照着本精读文档进行理解。文章标注的单词为重点单词，只提供相关注释，想要详细释义麻烦同学查一下词典哦！另外文章底色为灰色的句子为今天的翻译作业，今日文档中的句子分析模块会提供答案，同学们请先思考再看句子分析～】

【Para. 1】

All around us, every day, hundreds of conversations are taking place in languages we do not understand. Birdsong is one you would probably recognise, and you may have heard of honeybee waggle dances. But what about the spider **tuning** their web to listen for **prey**, or a **waft** of arousing perfumes **emitted** by a passing beetle?

- **insect** n. /'ɪnsekt/ any small creature with six legs and a body divided into three parts. Insects usually also have wings. Ants, bees and flies are all insects 昆虫

insect species 昆虫种类

- **tune** v. /tju:n/ to adjust a musical instrument so that it plays at the correct pitch (为乐器) 调音, 校音; 此处引申为"调节" 【考研大纲词汇】

Smart businesses always tune in to the needs of their customers. 聪明的企业总是能敏锐地关注到客户的需求。

- **prey** n. /preɪ/ an animal, bird etc. that is hunted and eaten by another animal 被捕食的动物, 猎物 【考研大纲

词汇】

The lion moved slowly towards its prey. 狮子缓缓向猎物逼近。

- **waft** n. /wɒft/ a smell or smoke that moves through the air (空气中飘荡的) 一阵气味, 一缕烟

A waft of perfume drifted across the room. 一阵香水味飘过房间。

- **emit** v. /iˈmit/ to send out something such as heat, light, sound, gas etc. 发出, 射出, 散发 (热、光、声、气体等) 【考研大纲词汇】

The machine emits a high-pitched signal when the water is ready. 水烧开时, 机器会发出一声尖锐的提示音。

在我们身边, 每天都上演着数百场我们听不懂的"对话"。鸟鸣或许是你能辨认的一种, 你也可能听说过蜜蜂的"摇摆舞"。但你可曾想过, 蜘蛛会通过调节蛛网来"听"辨猎物, 又或者, 一只路过的甲虫会散发出一阵撩人的"香水"?

【Para. 2】

To **delve** into the world of **invertebrate** communication about food gathering, nest building, defending a **territory**, attracting a mate, and caring for their young is to realise just how smart and

sophisticated they are.

- **delve** v. /delv/ ~ (into sth) to try hard to find out more information about something 探索，探究，深入研究

The report delves into the social impacts of artificial intelligence. 该报告深入探究了人工智能的社会影响。

- **invertebrate** n. /ɪn 'vɜ:tɪbrət/ an animal that does not have a backbone 无脊椎动物

Insects, spiders and worms are all invertebrates. 昆虫、蜘蛛和蠕虫都属于无脊椎动物。

- **territory** n. / 'terətri/ an area that an animal or person regards as its own and defends against others 领地，领土，地盘【考研大纲词汇】

The wolf marks its territory by leaving scent marks on trees. 狼通过在树上留下气味标记来划定领地。

深入无脊椎动物的交流世界——无论是觅食、筑巢、守卫领地、吸引配偶，还是抚育后代——你都会意识到，它们是何等聪慧与精妙。

【Para. 3】

Spiders, for example, can “hear” **vibrations** on the silken **strands** of their web using sensors on their legs. With their legs and

body weight, the spiders will adjust the silk strings, as we would a guitar string to change its **pitch**, pulling, loosening or reconnecting strands to adjust how it sounds. In fine-tuning the silk they can know just who or what is on it.

- **vibration** n. /vaɪˈbreɪʃn/ a continuous slight shaking movement 震动，振动，颤动【考研大纲词汇】

The vibration of the engine could be felt throughout the building. 整栋楼都能感觉到发动机的震动。

- **strand** n. /strænd/ a single thin piece of thread, wire, hair etc. (线、绳、发等的) 股，缕

She tucked a strand of hair behind her ear. 她把一缕头发别到耳后。

- **pitch** n. /pɪtʃ/ how high or low a sound is, especially a musical note (声音或音符的) 高低，音高【考研大纲词汇】

Her voice rose to a high pitch as she became more excited. 她越激动，嗓音就越尖。

以蜘蛛为例，它们能利用腿部的传感器"听"到蛛丝上的振动。蜘蛛会用腿和体重来调节蛛丝，就像我们调吉他弦以改变音高一样——通过拉紧、放松或重新连接蛛丝来调整它的"音色"。经过对蛛丝的精细调校，蜘蛛便能判断出落在网上的究竟是谁、是什么。

【Para. 4】

Many invertebrates use **pheromones** – complex ranges of **scents** – to communicate. Moths can mix a **cocktail** of perfumes to attract a mate while ants lay chemical **trails** to guide their sisters to food sources. Some burying beetles share parenting and will communicate using chemical scents, as well as touching.

- **pheromone** n. / 'ferəməʊn/ a chemical substance produced by an animal that influences the behaviour of others of its species 信息素，外激素

Ants use pheromones to mark trails leading to food.

蚂蚁用信息素标记通往食物的路径。

- **scent** n. /sent/ a pleasant smell; the smell that a particular animal or person has 香味，气味；（动物的）臭迹，遗臭【考研大纲词汇】

The scent of roses filled the garden.

花园里弥漫着玫瑰的芬芳。

- **cocktail** n. / 'kɒkteɪl/ a mixture of different things, often an unpleasant one（常指不好的）混合物，综合体；鸡尾酒

The city offers a heady cocktail of culture and

commerce.

这座城市融合了文化与商业，令人沉醉。

- **trail n.** /treil/ a long line or mark that is left behind by somebody/something as they move 踪迹，痕迹，痕迹路径【考研大纲词汇】

The hurricane left a trail of destruction across the coast. 飓风在沿海地区留下了满目疮痍。

许多无脊椎动物利用信息素——一系列复杂的气味——来交流。飞蛾能调配出"香水鸡尾酒"以吸引配偶，而蚂蚁则铺设化学路径，引导同伴前往食物来源。一些埋葬甲虫会共同抚育后代，除了肢体接触外，也通过化学气味进行交流。

【Para. 5】

Mimicry is another tool used to deceive. The hoverfly that mimics the yellow and black stripes of the wasp, to **deter predators** such as birds from eating them, is well known. Spiders and moths can mimic ants in shape, smell and movement so they are able to live and eat in an ant nest rent-free. Several insects mimic parts of a tree to avoid being eaten. Others, such as some praying mantids, can appear perfectly like the flowers they are waiting in, all the better to lie undetected and **pounce** on unsuspecting prey.

- **mimicry** n. / 'mɪmɪkri/ the act of copying the way somebody/something speaks, behaves, looks etc. 模仿, 模拟; (生物) 拟态

The comedian's mimicry of the politician was spot on.
那位喜剧演员对政客的模仿惟妙惟肖。

- **deter** v. /dɪ 'tɜ:(r)/ to make somebody decide not to do something, especially by making them realize the difficulties 阻止, 威慑, 使不敢【考研大纲词汇】

The high cost of the procedure deters many patients from seeking treatment. 高昂的手术费用让许多患者不敢就医。

- **predator** n. / 'predətə(r)/ an animal that kills and eats other animals 捕食者, 食肉动物; 掠夺者【考研大纲词汇】

The cheetah is one of the fastest predators on the savannah. 猎豹是大草原上速度最快的捕食者之一。

- **pounce** v. /paʊns/ ~ (on sb/sth) to move forward suddenly in order to catch or attack somebody/something 猛扑, 突袭

The cat crouched and then pounced on the mouse. 猫蹲伏下来, 然后猛地扑向老鼠。

拟态是另一种用于欺骗的交流工具。食蚜蝇模仿黄蜂的黄

黑条纹以吓退鸟类等捕食者，这已是众所周知的例子。蜘蛛和飞蛾能在外形、气味和行动方式上模仿蚂蚁，从而在蚁巢中不付“房租”地白吃白住。还有一些昆虫会模仿树木的局部以避免被吃掉。另一些昆虫，比如某些螳螂，能完美地伪装成它们藏身其中的花朵，以便潜伏不被发现，然后对毫无防备的猎物发起突袭。

【Para. 6】

One butterfly **caterpillar** uses an incredible two-tone “song” to live undetected in an ant nest. Having **tricked** the ants, by mimicking ant baby smells, into carrying them into the nest; once **installed**, the caterpillar will make noises so similar to that of an ant **larvae** that they get fed and protected by unsuspecting ant workers.

- **caterpillar** n. /'kætəpɪlə(r)/ a small creature like a worm with many legs, that develops into a butterfly or moth
毛虫，幼虫（蝴蝶或蛾子的幼体）

The caterpillar will eventually form a chrysalis and emerge as a butterfly.

毛虫最终会结蛹，然后羽化为蝴蝶。

- **trick** v. /trɪk/ to make somebody believe something that is not true, especially in order to get what you

want 欺骗，哄骗，诱骗【考研大纲词汇】

He tricked me into lending him my car.

他骗我把车借给了他。

- **install v.** /ɪnˈstɔ:l/ to put somebody/something in a particular place or position, often officially 安置，使就职；安装【考研大纲词汇】

She was installed as the new director last month. 她上个月正式就任新主管。

- **larvae n.** /ˈlɑ:vi:/ (plural of larva /ˈlɑ:və/) a young form of an insect that has just come out of an egg and looks like a short fat worm 幼虫，幼体

The mosquito larvae live in water before they develop into adults.

蚊子幼虫在发育为成虫之前生活在水中。

有一种蝴蝶幼虫会用一种令人惊叹的双音调"歌声"在蚁巢中隐身生存。它先通过模仿蚁宝宝的气味骗得蚂蚁把自己搬入巢中；一旦安顿下来，幼虫便会发出与蚁幼虫极为相似的声音，从而得到毫无戒心的工蚁的喂食和保护。

【Para. 7】

And it's not just invertebrate talking to invertebrate. Trees, for

example, communicate with bees, using colour and **ultraviolet** light to attract and guide them to **pollinate** flowers.

- **ultraviolet** adj. /ˌʌltrəˈvaɪələt/ (of light) that is beyond the violet end of the range of colours that people can see
紫外线的，紫外的

Too much ultraviolet radiation can cause skin cancer.

过多的紫外线辐射可能导致皮肤癌。

- **pollinate** v. /ˈpɒləneɪt/ to put pollen into a flower or plant so that it produces seeds 给……授粉，传粉

Bees pollinate many of the crops we rely on for food.

蜜蜂为许多我们赖以生存的农作物授粉。

而且，交流并不只发生在无脊椎动物之间。例如，树木会与蜜蜂交流——利用颜色和紫外线吸引并引导蜜蜂为花朵授粉。

【Para. 8】

Some plants **beckon** bugs, such as crab spiders, to attack other bugs which are attacking them, while the relationship between the fig tree and the fig wasp is particularly **intimate**. Every type of fig tree has a unique perfume to call its own wasp species. The tiny female wasp crawls through a hole in the fig flower, often having to lose her wings to get through, and brings pollen from another tree

with her. A whole wasp life cycle will then occur inside what you will eventually eat as a juicy fig.

- **beckon v.** / 'bekən/ to call or signal to somebody with a movement of the hand or head; to appear very attractive 招手示意，召唤；吸引，诱惑

The opportunity to work abroad beckoned her.

出国工作的机会在向她招手。

- **intimate adj.** / 'intɪmət/ having a very close relationship; very detailed and thorough 亲密的，密切的；详尽的，深入的【考研大纲词汇】

The two countries have established intimate diplomatic ties. 两国建立了密切的外交关系。

有些植物会招来蟹蛛等昆虫，让它们攻击侵害自己的其他虫子；而无花果树与榕小蜂之间的关系则尤为亲密。每一种无花果树都有独特的“香水”来召唤专属的榕小蜂物种。微小的雌蜂爬过榕花上的一个小孔——往往不得不折断翅膀才能通过——并随身带来另一棵树的花粉。随后，一整个榕小蜂的生命周期便在你最终将作为多汁无花果食用的果实内部完成。

【Para. 9】

Invertebrates have **evolved** these relationships over millions of

years, finding ways to inhabit their **niche** better in collaboration with each other and the rest of the world. And it is not all about cooperation. Much communication – from pugilistic ants to worms biting each other – is about status, **repelling** predators, winning mates and protecting territory, babies or food.

- **evolve v.** /i'vɒlv/ to develop gradually, especially from a simple to a more complex form 进化，演变，逐步发展
【考研大纲词汇】

Humans evolved from earlier primate species over millions of years. 人类在数百万年间从更早的灵长类物种进化而来。

- **niche n.** /ni:ʃ/ a comfortable or suitable position in life, work, or society; the role of an organism within its environment 生态位；合适的位置，舒适的定位

The company has found its niche in the luxury market.
这家公司在奢侈品市场找到了自己的定位。

- **repel v.** /rɪ'pel/ to drive or force somebody/something back; to make somebody feel disgusted 击退，驱逐；使厌恶，使反感【考研大纲词汇】

The army repelled the enemy attack and defended the city. 军队击退了敌军的进攻，守住了城池。

无脊椎动物在数百万年间进化出了这些关系，通过彼此以

及与世界其他物种的协作，找到了更好地占据自身生态位的方式。但这一切并不全是合作。大量的交流——从好斗的蚂蚁到互相撕咬的蠕虫——关乎地位、驱赶捕食者、赢得配偶，以及保护领地、幼崽或食物。

【Para. 10】

Scientists are still discovering the incredible detail of insect communication, without which **chaos** would **ensue**. Sadly, when humans interrupt these conversations, whether by **interfering** with their **habitat**, using chemicals or causing climate emergency, we cause serious problems – for ourselves as well as the wildlife.

- **chaos** n. /'keɪɒs/ a state of complete confusion and lack of order 混乱，紊乱，混沌【考研大纲词汇】

The sudden power cut plunged the office into chaos.

突然停电让办公室陷入一片混乱。

- **ensue** v. /ɪn'sjuː/ to happen after or as a result of something 继而发生，因而产生，接着发生【考研大纲词汇】

A heated debate ensued after the proposal was announced.

提案公布后随即引发了激烈的辩论。

- **interfere** v. / ,ɪntə 'fiə(r)/ ~ (with/in sth) to deliberately get involved in a situation where you are not wanted or needed 干涉，干预，介入【考研大纲词汇】

The government should not interfere in the internal affairs of other countries. 政府不应干涉他国内政。

- **habitat** n. / 'hæbitæt/ the natural home of a plant or animal 栖息地，生活环境【考研大纲词汇】

The destruction of forests threatens the habitat of countless species. 森林的破坏威胁着无数物种的栖息地。

科学家们仍在不断发现昆虫交流中令人惊叹的细节——没有这些交流，混乱便会接踵而至。遗憾的是，当人类打断这些"对话"时——无论是通过破坏它们的栖息地、使用化学药剂，还是引发气候危机——我们都在制造严重的问题，既危害野生动物，也危害我们自己。

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➤ 今日翻译作业

(Para. 3) With their legs and body weight, the spiders will adjust the silk strings, as we would a guitar string to change its pitch, pulling, loosening or reconnecting strands to adjust how it sounds.

(下一页提供句子分析)

➤ 今日句子分析

Para 3. With their legs and body weight, the spiders will adjust the silk strings, as we would a guitar string to change its pitch, pulling, loosening or reconnecting strands to adjust how it sounds. 蜘蛛会用腿和体重来调节蛛丝，就像我们调吉他弦以改变音高一样——通过拉紧、放松或重新连接蛛丝来调整它的"音色"。

主句

With their legs and body weight, the spiders will adjust

方式状语

主语

谓语

方式状语从句（省略结构）

the silk strings, as we would [adjust] a guitar string

宾语

连词

主语

助动词

（省略谓语 adjust）

宾语

to change its pitch, pulling, loosening or reconnecting

目的状语

现在分词短语作伴随状语

strands to adjust how it sounds.

目的状语（含宾语从句）

在主句中：

方式状语：With their legs and body weight;

主语：the spiders; 谓语：will adjust;

宾语：the silk strings;

在方式状语从句中：

连词：as；

主语：we； 助动词：would；

宾语：a guitar string；

目的状语：to change its pitch；

现在分词短语作伴随状语：pulling, loosening or reconnecting strands；

目的状语：to adjust how it sounds.

本句主干为 the spiders will adjust the silk strings（蜘蛛会调节蛛丝），整体结构清晰，但包含三个层次的修饰成分。句首 With their legs and body weight 为介词短语作方式状语，说明蜘蛛用什么来调节；as we would a guitar string to change its pitch 为方式状语从句，其中省略了与主句相同的谓语动词 adjust，完整形式应为 as we would adjust a guitar string，to change its pitch 为目的状语；句末 pulling, loosening or reconnecting strands 为三个并列的现在分词短语作伴随状语，修饰主句 the spiders 和谓语 adjust，表示调节的具体动作，to adjust how it sounds 为目的状语，其中 how it sounds 为宾语从句。翻译时应注意 as 从句的省略结构，将"正如我们调吉他弦以改变音高一样"这一比喻自然融入中文表达，并将三个现在分词短语译为并列的动作描述。

➤ 写作积累

Para 10. Scientists are still discovering the incredible detail of insect communication, **without which** chaos would ensue.

- ♦ without which —— "没有了它/它们", 介词+关系代词引导非限制性定语从句。which 指代前文提到的事物（此处为 insect communication），without 表示"如果没有"。这一结构相当于 if there were no..., 常与虚拟语气连用（本句中 chaos would ensue 即为虚拟语气），表达一种与现实相反的假设。在写作中，without which 能够简洁有力地强调某事物的不可或缺性，比使用 if-clause 更加紧凑和书面化。例如：The internet has become an essential part of modern life, without which many daily activities would grind to a halt.（互联网已成为现代生活不可或缺的一部分，没有它，许多日常活动将陷入停滞。）
- ♦ ensue 是正式用语，表示"继而发生、随之而来"，chaos would ensue 是一个极具画面感的表达，强调一旦失去某种关键要素，混乱将不可避免地随之而来。相比 chaos would happen 或 there would be chaos, ensue 更强调因果关系和时间上的承接，常用于学术写作和正式文体中。例如：If the government fails to act, a humanitarian crisis would ensue.（如果政府不采取行动，人道主义危机

便会接踵而至。)